

REQUEST FOR INFORMATION (RFI): Atmospheric Microwave Radiometry Observations for NASA FALCON

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Short URL: <https://go.nasa.gov/atmrforfi>

The National Aeronautics and Space Administration (NASA) Science Mission Directorate (SMD) is soliciting information on low-cost concepts for the development, launch, and operations of a radiometer component to fly in formation with the Fleet for the Atmosphere Linking Commercial Observations with NASA (FALCON). For this Request for Information (RFI), NASA is targeting a commercially owned and operated microwave radiometer capability to fly in formation with the NASA constellation. The instrument, supporting platform, launch, and operations should fully leverage commercial capabilities, and this RFI invites recommendations on the most effective approaches to facilitate this.

This RFI seeks responses broadly from industry, universities, non-profit organizations, individuals, NASA centers, Federally Funded Research and Development Centers (FFRDCs), University Affiliated Research Centers (UARC)s, other U.S. federal, state, local, and tribal government agencies, and the international community. NOTE: Responses are subject to the restrictions placed on NASA by Section 526 of the “Consolidated Appropriations Act, 2024” and all applicable subsequent Appropriations Acts that prohibit bilateral relationships with China. Details of the restriction are at: <https://science.nasa.gov/researchers/sara/faqs/prc-faq-roses/>

1. Background

The following information should be used by the responder to understand formation flying with FALCON and driving requirements for the instrument.

FALCON Information

FALCON is the NASA Earth Science Division’s (ESDs) response to the atmospheric components of the 2017 National Academies of Science, Engineering, and Medicine Earth Science Decadal Survey entitled, “*Thriving on Our Changing Planet: A Decadal Strategy for Earth Observations from Space.*” FALCON directly addresses the scientific elements of the Clouds, Convection, and Precipitation and Aerosols targeted observables. FALCON also will incorporate relevant explorer- and venture-class missions with an aim towards integrated science and applications across both the atmosphere and the Earth system at large, as laid out in the [Earth Science to Action Strategy 2024-2034](https://assets.science.nasa.gov/content/dam/science/esd/earth-science-division/earth-science-to-action/ES2A_Booklet_web.pdf) (https://assets.science.nasa.gov/content/dam/science/esd/earth-science-division/earth-science-to-action/ES2A_Booklet_web.pdf).

FALCON is a fleet of missions that will begin with the launch of NASA's existing INvestigation of Convective UpdraftS (INCUS) and Polarized Submillimeter Ice-Cloud Radiometer (PoLSIR) venture-class missions and continue to grow through the FALCON Lidar, FALCON Radar, and existing explorer-class Stratosphere Troposphere Response using Infrared Vertically-resolved light Explorer (STRIVE) mission, as well as through international collaborations and partnerships. FALCON science will be conducted according to NASA's Open Science policy.

As the focus of this RFI, NASA is investigating commercial options for adding passive microwave radiometry to the constellation. Concepts shall be executable and realistically launch by Quarter 3 Fiscal Year 2029.

FALCON Radiometer Satellite Information

Since the orbital architecture of the FALCON constellation is still under investigation, NASA is seeking responses that envelop a range of possibilities as the FALCON constellation architecture is refined. Ultimately, it is expected that the FALCON Radiometer concept will fly in coordination with other FALCON satellites and will not individually drive the orbital parameters.

Below is a list of key orbital parameters that should be considered as part of the submissions. In the event that specific parameters are listed, they should be considered as an envelope of the trade space to be considered in the responses:

- Altitude Range: 400 km – 550 km
- Inclination: Both traditional sun-synchronous orbits and mid-inclination orbits ~55 deg
- Local Time of Ascending Node or Right Ascension of the Ascending Node, if limited
- Duty cycles and modes of operations, if limited
- Ability to station keep on orbit
- Maturity of capabilities

As part of the concepts, submissions are asked to quantify any limitations imposed on satellite operations as a function of the parameter space defined above.

FALCON Radiometer Instrument Information

Passive microwave radiometers provide vital information on surface and atmospheric phenomena across the Earth system. These observations measure emitted radiation from the Earth and its atmosphere, and these observations are further modified by the absorption and scattering by the composition of the atmosphere itself. In the microwave, these observations are sensitive to clouds and precipitation within the system. It is this sensitivity that will be fundamental in adding important microphysical information and spatial contextualization to the FALCON constellation.

As part of the submission, concepts shall address the following instrument parameters.

- Spectral capabilities in the millimeter and sub-millimeter wave bands (e.g., number of channels; spectral range, bandwidth and resolution; radiometric accuracy and noise performance).
- Scanning and viewing geometry (e.g., cross- vs. conical-scanning, spatial resolution characterization, sampling characteristics).
- Onboard calibration methods and expected stability.
- Data delivery metrics (e.g., data volume, data latency).
- Maturity of capabilities.

Submissions shall quantify any dependencies or limitations imposed on instrument observations and operations as a function of the satellite parameter space defined above.

2. Requested Information

Respondents may not submit confidential information, controlled unclassified information (CUI), proprietary information, or export-controlled information, including International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR) restricted information, in response to this RFI.

This RFI specifically seeks information on the reformulated constellation of NASA Earth observation mission as detailed above. For new missions, NASA is especially interested in approaches that increase efficiency, reduce time to science, leverage commercial capabilities, encourage partnerships, and to rely more on industry for the end-to-end mission execution. NASA is open to all proposals and novel ideas for productive arrangements, including concepts that leverage capabilities or incorporate best practices from commercial fleet operators and innovative partnering opportunities that may help achieve greater science objectives.

Arrangements could include, but are not limited to

- Data buy only
- Co-funding technology development for the commercial mission early in the cycle and then buying data as a service
- Creating capability for a fully commercial venture demonstration

The response shall be no longer than 10 pages and include, at a minimum, the following:

a. Organization Information:

1. Description of organization or consortium of organizations, including any partnerships. As part of a partnership, identify any role that NASA can contribute to help meet or exceed requirements (e.g., NASA contributions to calibration, dependencies on current NASA assets for calibration).
2. Description of organizational experience and capabilities pertinent to performing the mission described in this RFI.

b. Mission Concept Approach(es): Description of the envisioned mission concept to accomplish the mission as outlined in Section 1, including:

1. Addressing the specific information in Section 1.

2. Description of proposed programmatic approaches and arrangements to help implement a commercial approach, including structure of the contract/agreement, management approaches, and performance/milestone incentives.
 3. Approaches to reduce time to science.
 4. Considerations for data aspects and data rights, including any restrictions on delay, use, and distribution, and licensing options.
- c. Cost and Schedule:
1. Rough Order of Magnitude (ROM) cost estimate in Government fiscal year 2026 dollars (FY26\$).
 2. Estimate of the timing and durations of major activities.
- d. Additional Information: Provide additional insights not captured above.

3. Response Instructions

All responses must be received by 11:59 p.m. Eastern time on April 23, 2026.

All responses to this RFI must be submitted in an electronic format via the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES), home page located at <http://nspires.nasaprs.com/>.

There is no limit on the number of responses from a particular submitter; that is, an individual, an institution, or organizational units at an institution. Individuals currently unaffiliated with any commercial, research, educational, or other institution may list their institutional affiliation as “Self”. Submissions from outside the United States are also welcome.

For this RFI, a response submission will take the form of a Notice of Intent (NOI) within the NSPIRES online announcement data management system. The RFI response itself will be a PDF-formatted document that is attached (uploaded) to NSPIRES. All responses must adhere to the following formatting requirements:

Length:	No more than 10 pages plus a title page
Page size:	8.5” x 11.0” paper size
Paragraph:	Single-spaced, single-column text with no more than 5.5 lines per vertical inch of text
Margins:	One-inch margins on all four sides with no content in the margins
Font:	Font size 12, not to exceed 15 characters per horizontal inch, including spaces, a sans serif font recommended.
Inserts:	Figures, tables, and other inserts are permitted

You must be registered with NSPIRES to submit a RFI response. See registration instructions at <http://nspires.nasaprs.com/> (select “Create an Account” at the top right of the landing page). Neither institution registration nor an institution affiliation is required to respond to this RFI.

When you have an NSPIRES account, go to the NSPIRES page for Request for Information: Radiometer for FALCON, Number NNH26ZDA007L: at <https://go.nasa.gov/atmrorf>

1. Click "Create"
2. Log in when requested
3. Enter the Request for Information: Response title
NOTE: NSPIRES will show this as "NOI title"
4. Select "do not link at this time" for submitting organization page.
5. Click "Save" on next page.
6. It is not necessary to complete any of the "NOI Details"; all requested information should be included in the attached PDF document. Information which is entered into "NOI Details" but not included in the attached PDF document will not be considered
7. Prepare your RFI response offline and save as a PDF document (NSPIRES instructions on PDF formats are described in the NSPIRES PDF Guidelines available at https://nspires.nasaprs.com/external/externalhelp/public/index.htm#t=NSPIRES_Documents_main.htm. The response document must include the respondent's Name, institution, and e-mail address so the file is self-contained. File names format should be "PI Last Name - First Name – Number - RFI". "Number" will be used to distinguish multiple responses from the same PI. Each response should not exceed 7 pages in length excluding references.
8. To attach (upload) your PDF document:
 - a.) Click "add" under NOI attachments section;
 - b.) Select "Proposal Document" from the drop-down list;
 - c.) Browse to attach your PDF file;
 - d.) Select "Upload";
 - e.) Click "OK";
 - f.) Your RFI document has been uploaded to NSPIRES.
9. Click Submit NOI button. NOTE: Clicking "Submit NOI" does not complete the submission process.
10. Ignore any warnings about incomplete NOI elements.
11. Click "Submit". This will take you to the NOI submission confirmation page, which provides you with the NOI/RFI number for your records.

Please note: You may delete and replace an uploaded document any time before the submission deadline; however, once your RFI is submitted, it cannot be deleted. Unsubmitted responses will not be considered.

This is a request for information for planning and information purposes only. This is not a request for proposal or quotation, nor is this a solicitation for a contract or grant award. This RFI does not obligate the Government in any way. The Government will not reimburse the respondents for any costs associated with the information submitted in response to this request.

No corresponding solicitation exists; therefore, do not request a copy of the solicitation. If a solicitation following this is released, it will be posted on www.sam.gov (<https://www.sam.gov>). It is the interested party's responsibility to monitor this site for the release of any solicitation or synopsis.

Responses to this synopsis will be reviewed by both civil servants and support service contractors.

4. Points of Contact

Please email questions and comments concerning this RFI to the email addresses below. Depending on the nature of received questions, NASA may respond on an individual basis by email or may post responses to inquiries in a “Questions and Answers (Q&A)” document available at <https://go.nasa.gov/atmrorf>. Any posted Q&As will be edited to preserve the anonymity of persons and institutions who submit questions and are intended to address inquiries of broader interest and general clarification.

The emails below are only for questions and comments. NASA will not consider any RFI submissions sent to these email addresses.

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